Faysal Hossain Shezan

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Appointment

Assistant Professor (tenure-track) at University of Texas at Arlington **September 2023 - present**

Education

PhD in Computer Science **August 2023**
Thesis: Cross-Platform Security and Privacy Analysis of Emerging Systems
Advisor: Dr. Yuan Tian
University of Virginia

Visiting Research Assistant **January 2023 - August 2023**
Host: Dr. Yuan Tian
University of California, Los Angeles

Masters in Computer Science **August 2022**
CGPA 4.00/4.00
University of Virginia

BSc in Computer Science and Engineering **March 2016**
Thesis: High Performance Approximate Computing by Adaptive Relaxed Synchronization
Thesis Advisor: Dr. Rifat Shahriar
Bangladesh University of Engineering and Technology

Research Overview

My research lies at the intersection of security & privacy and machine learning, centered on one question: does a system do what it claims? I build automated techniques that measure, attack, and repair the gap between claimed and actual behavior, pairing LLMs with checkable machinery such as solvers, static analysis, and execution sandboxes. My work spans three areas: trustworthy AI-driven software and agents, including secure and copyright-aware code generation, exploit validation, and agent skills and memory; privacy and regulatory compliance, from GDPR and CCPA violation detection to verified repair and EU AI Act audits; and the security of emerging AI representations and sensing, including vision-language models, 3D Gaussian Splatting, generative audio, and wearable and XR authentication.

Publications

- [**NeurIPS'26**]: Md Mahedi Hasan Rigan, Nicole Meng, Miao Yin, Yingjie Lao, **F. H. Shezan**, MVPISplat: Multi-View Photometric Inconsistency for Defending 3D Gaussian Splatting Attacks, The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS), 2026[pdf]
- [**EMNLP'26**]: Rui Chu, Nicole Meng, Tanusree Das Tithy, Thomas Cao, **F. H. Shezan**, Yingjie Lao, AegisCodeGen: Toward Copyright-Aware Code Generation with Runnability-Guided Fusion, The 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026 [pdf]
- [**ISSTA'26**]: Tanusree Das Tithy, Lamia Hasan Rodoshi, Ayman Rafid Azahar, Amlan Abhidarshi, Tabassum Faruk, Fahmid Al Rifat, **F. H. Shezan**, PoCE: Automated Proof-of-Concept Synthesis using Large Language Models for Robust Exploit Validation, Proceedings of The 35th ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), 2026 [pdf]
- [**ICML'26**]: Nicole Meng, Ronak Sahu, Miao Yin, **F. H. Shezan**, Yingjie Lao, "WhisperSplat: Lossless Steganography in 3D Gaussian Splatting.", Proceedings of The 43rd International Conference on Machine Learning (ICML), 2026 [pdf]
- [**AsiaCCS'26**]: Tanusree Das Tithy*, Poojitha Thota*, Shirin Nilizadeh, **F. H. Shezan**, "The Insider's Advantage: Exploiting Automated Privacy Policy Analyzer Tools Through Subtle Text Manipulations.", Proceedings of The 21st ACM ASIA Conference on Computer and Communications Security, 2026 [pdf] (* co-first author)
- [**IEEE S&P'26**]: Ying Li, Wenjun Qiu, **F. H. Shezan**, Kunlin Cai, Michelangelo van Dam, Lisa Austin, David Lie, Yuan Tian, "Breaking the illusion: Automated Reasoning of GDPR Consent Violations.", Proceedings of The 47th IEEE Symposium on Security and Privacy, 2026 [pdf]
- [**USENIX'25**]: Ying Li, **F. H. Shezan**, Bomin Wei, Gang Wang, Yuan Tian, "SoK: Towards Effective Automated Vulnerability Repair", Proceedings of The 34th USENIX Security Symposium (USENIX Security), 2025 [pdf]
- [**SOUPS'23**]: M. McCall, E. Zeng, **F. H. Shezan**, M. Yang, L. Bauer, A. Bichhawat, C. Cobb, L. Jia, Y. Tian, "Towards Usable Security Analysis Tools for Trigger-Action Programming". Symposium on Usable Privacy and Security 2023 [pdf]
- [**PETS'23**]: **F. H. Shezan**, M. Long, D. Hasani, G. Wang, Y. Tian, "SenRev: Measurement of Personal Information Disclosure in Online Health Communities", Proceedings of the Privacy Enhancing Technologies Symposium 2023 [pdf]
- [**NDSS'23**]: **F. H. Shezan**, Z. Su, M. Kang, N. Phair, P. W. Thomas, M.V. Dam, Y. Cao, Y. Tian, "CHKPLUG: Checking GDPR Compliance of WordPress Plugins via Cross-language Code Property Graph", In Network and Distributed Systems Security (NDSS) Symposium 2023 [pdf]

- [**IEEE CNS'22**]: **F. H. Shezan**, Y. Lao, M. Peng, X. Wang, M. Sun, P. Li, "NL2GDPR: Automatically Develop GDPR Compliant Android Application Features from Natural Language", IEEE Conference on Communications and Network Security (IEEE CNS 2022) [pdf]
- [**IMWUT/UbiComp'21**]: **F. H. Shezan**, H. Hu, G. Wang, Y. Tian, "VerHealth: Vetting Medical Voice Applications through Policy Enforcement", Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies 2021 [pdf]
- [**WWW'20**]: **F. H. Shezan**, H. Hu, J. Wang, G. Wang, Y. Tian, "Read Between the Lines: An Empirical Measurement of Sensitive Applications of Voice Personal Assistant Systems", Proceedings of the Web Conference 2020 [pdf]
- [**NDSS'20**]: **F. H. Shezan**, K. Cheng, Z. Zhang, Y. Cao, Y. Tian, "TKPERM: Cross-platform Permission Knowledge Transfer to Detect Overprivileged Third-party Applications", In Network and Distributed Systems Security (NDSS) Symposium 2020 [pdf]
- [**IMWUT/UbiComp'20**]: Y. Lee, Y. Zhao, J. Zeng, K. Lee, N. Zhang, **F. H. Shezan**, Y. Tian, K. Chen, X. Wang, "Using sonar for liveness detection to protect smart speakers against remote attackers", Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies 2020 [pdf]
- [**NSysS'17**]: **F. H. Shezan**, S. F. Afroze, A. Iqbal, "Vulnerability Detection in Recent Android Apps: An Empirical Study", Proceedings of International Conference on Networking, Systems and Security 2017 [pdf]
- [**ITID'17**]: I. Ahmed, S. Guha, M. R. Rifat, **F. H. Shezan**, N. Dell, "Privacy Vulnerabilities in the Practices of Repairing Broken Digital Artifacts in Bangladesh", Information Technologies and International Development 2017 [pdf] (***best of ICTD paper***)
- [**HPCC'16**]: B. Islam, **F. H. Shezan**, R. Shahriyar, "High Performance Approximate Computing by Adaptive Relaxed Synchronization", Proceedings of International Conference on High Performance Computing and Communications 2016 [pdf]
- [**ICTD'16**]: I. Ahmed, S. Guha, M. R. Rifat, **F. H. Shezan**, N. Dell, "Privacy in Repair: An Analysis of the Privacy Challenges Surrounding Broken Digital Artifacts in Bangladesh", Proceedings of International Conference on Information and Communication Technologies and Development 2016 [pdf]
- [**IEEE WIECON-ECE'15**]: S. S. Rahman, S.S. Nusaka, **F. H. Shezan**, M. A. R. Sarkar, "The Development of Low Cost Exercise Monitoring Device for Paralytic", IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering 2015 [pdf]

Workshop Papers

- [**XR Security'26**]: N. Ntokos*, N. K. Dey*, G. S. Siripurapu*, W. Luo, S. T. L. Do, **F. H. Shezan** and J. Meng, "Meta Lens: A Runtime Measurement Framework for Meta Quest Camera Security", Co-located with ACM MobiCom 2026 [pdf]
- [**SPAR-3D'26**]: Md Mahedi Hasan Rigan, Nicole Meng, Miao Yin, Yingjie Lao, **F. H. Shezan**, "Adversarial Defense in Gaussian Splatting Using 3D Gaussian Filtering", SPAR-3D: First Workshop on Security, Privacy, and Adversarial Robustness in 3D Generative Vision Models, Co-located with CVPR 2026 [pdf]
- [**SSPXR'22**]: Z. Su, **F. H. Shezan**, Y. Tian, D. Evans and S. Heo, "Perception Hacking for 2D Cursorjacking in Virtual Reality", Proceedings of Safety, Security and Privacy in Extended Reality, Co-located with CHI 2022 [pdf]
- [**MASSW'19**]: Y. Yu, C. Li, M. A. Jonas, C. Ma, **F. H. Shezan**, S. Shen, P. Gao, Y. Tian "Detecting Abnormal Behaviors in Smart Home", International Conference on Mobile Ad Hoc and Sensor Systems Workshops 2019 [pdf]
- [**SPW'19**]: S. Liu, Y. Wei, J. Chi, **F. H. Shezan**, and Y. Tian, "Side Channel Attacks in Computation Offloading Systems with GPU Virtualization", IEEE Security and Privacy Workshops 2019, [pdf]

Posters

- [**ACM MobiCom'24**]: "Poster: Do Privacy-Preserving Obfuscation Techniques Degrade the Accuracy of Odometry?", N. Ntokos, N. K. Dey, J. Meng, and **F. H. Shezan**, ACM MobiCom 2024 [pdf]
- [**IEEE S&P'24**]: "Poster: Enhancing Symbolic Execution with LLMs for Vulnerability Detection", M. N. Zaharudin, M. H. Zuhaimi, **F. H. Shezan**, IEEE Symposium on Security and Privacy 2024 [pdf]
- [**IEEE S&P'24**]: "Poster: Identifying Copyright Code in Training Corpus of Large Language Model", L. H. Rodoshi, F. Tafannum, **F. H. Shezan**, IEEE Symposium on Security and Privacy 2024 [pdf]
- [**MobiSys'17**]: "Poster: HeartFit: An Intuitive Smartphone Application for Well-being of Hypertensive Patients", S. F. Afroze, **F. H. Shezan**, and S. Sharmin, Proceedings of the annual international conference on mobile systems, applications, and services 2017 [pdf]

Honors and Awards

H13. Roots of Dedication-Service Award, UTA
H12. Travel grant from IEEE S&P (2023), NDSS (2023), IEEE CNS (2022), WWW (2020)
H11. UVA Endowed Graduate Fellowship Award (Copenhaver Charitable Trust Bicentennial Fellowship), This award goes to a graduate student who has exceptionally good academic performance, research productivity (publications, presentations), and awards/honors, November 2022
H10. Selected to participate in NRT Graduate Student Communication Research Series, August 2022
H9. Selected as CPS Rising Stars, April 2022
H8. UVA Endowed Graduate Fellowship Award (Carlos and Esther Farrar Fellowship), This award goes to a graduate student who has exceptionally good academic performance, research productivity (publications, presentations), and awards/honors, October 2021
H7. Link Lab Outstanding Graduate Research Award, This award goes to graduate student who has demonstrated

excellence in research during the academic year, June 2020
H6. Blackhat Grant, Registration grant from BlackHat USA, August 2018
H5. Bangladesh Sweden Trust Fund Scholarship, July 2019
H4. PhD Fellowship, University of Virginia, March 2017
H3. Runner-up at *Mobile App Hackathon: Code Hub* by HEQEP, March 2014
H2. Runner-up at *Dev-Mercenaries System Prototyping & Development* by Department of CSE, BUET, March 2013
H1. Top WP App at *Windows App Hackathon* by Microsoft Bangladesh, March 2012

Teaching Experience

Teaching

- CSE 6392: Advanced Topic: Seminar in LLM Security, Privacy & Ethics, Fall 2026
- CSE 4380/5380: Information Security I, Fall 2025, 2024, 2023
- CSE 4380: Information Security I, Spring 2026, 2025, 2024
- CSE 4382/5382: Secure Programming, Spring 2026

Mentoring Experience

PhD Students
Nahin Kumar Dey, University of Texas at Arlington (January, 2024 - Present)
Tanusree Das Tithy, University of Texas at Arlington (June, 2024 - Present)
Fahmid Al Rifat, University of Texas at Arlington (August, 2025 - Present)
Promit Basak, University of Texas at Arlington (August, 2025 - Present)
Mahedi Hasan Rigan, University of Texas at Arlington (August, 2025 - Present)
Master Students
Lamia Hasan Rodoshi, University of Texas at Arlington (Graduated May, 2026)
Undergraduate Students
Marzuque Mashrur Fariz, University of Texas at Arlington (Jan, 2025- present)
Abigia Abate, University of Texas at Arlington (August, 2024- present)
Ayman Rafid Azahar, University of Texas at Arlington (June, 2024- present)
Joshua Olaoye, San Diego State University (CAHSI LREU Program, February, 2025 - May, 2025)
Makayla S Hall, University of Houston (CAHSI LREU Program, August, 2024 - June, 2025)

- second place in poster competition at GMIS Conference, 2025

David Temitope Awe, University of Texas at Arlington (August, 2024- Dec, 2025)
Bardan Dhakal, University of Texas at Arlington (June, 2024 - May, 2025)
Natalia Mesta Valenzuela, University of Texas at El Paso (CAHSI LREU Program, February, 2024 - May, 2024)
Muhammad Haziq Mirza Zuhaimi, University of Texas at Arlington (January, 2024 - May, 2024)
Muhammad Zaharudin, University of Texas at Arlington. (September, 2023 - May, 2024)

Professional Services

INSIDE UTA

- Organized Workshop Session at OurCS@DFW (February, 2024 and February, 2025)
- PhD Admission Committee (September, 2023-present)
- Colloquia Committee (September, 2023-present)
- Judge at HackUTA (October, 2023)
- Judge at PhD Lightening Talks During 50th Anniversary of CSE Department (October, 2023)

OUTSIDE UTA

Proposal Reviewer

- NSF EDU CyberAI SFS INNOVATION panel, 2026
- Coastal Virginia Center for Cyber Innovation (COVA CCI), 2026
- NSF Secure and Trustworthy Cyberspace (SaTC) review panel, 2023
- NSF Computer and Information Science and Engineering Research Initiation Initiative (CRII) review panel, 2023

Workshop Program Organizer
SPAR-3D: Security, Privacy, and Adversarial Robustness in 3D Generative Vision Models, co-located with CVPR 2026.
TPC
IEEE S&P (2027, 2026, 2025, 2024), USENIX (2027, 2026, 2025), NDSS (2027, 2026, 2025), CCS (2026, 2025, 2024), Texas Security Workshop (2026), The Web Conference (2025, 2024), RAID (2025), SecDev (2025), ICWSM (2024, 2023), Sensors S&P (2023), SafeThings (2023), ICICS (2023), CCS (Poster, 2022), IMC (Shadow TPC, 2022), USENIX Security (Poster, 2018)

Journal Reviewer
ACM Transactions on Privacy and Security (2025, 2026)
Session Chair
IEEE S&P 2023, 2024; NDSS 2023; CCS 2022; WPES 2022
Web Chair
SafeThings, 2023
External Reviewer
PoPETs (2022), IMWUT/Ubicomp (2022, 2024), ICWSM (2022), NordiCHI (2022), MobileHCI (Poster, 2022)
Sub-Reviewer
CCS (2018, 2019), IEEE S&P (2020, 2021, 2022), NDSS (2018, 2020, 2021, 2023), USENIX (2018, 2020), ASIACCS (2020), AAAI (2019), SENSYS (2020, 2021, 2022), CSF (2021), CHI (2021)
Batch Representative of BSADD (BUET Systems Analysis, Design & Development) [link] **(2015 - 2016)**

Featured

F2. CPS Rising Stars 2022 [Article]

F1. Humans of Link Lab [Article]

Talks

T9. NDSS, CHKPLUG: Checking GDPR Compliance of WordPress Plugins via Cross-language Code Property Graph, February 2023

T8. IEEE CNS, NL2GDPR: Automatically Develop GDPR Compliant Android Application Features from Natural Language, October 2022

T7. Guest Lecturer, Security and Privacy analysis for voice personal assistant. Course: CS 8501: IoT Security and Privacy, February 2022

T6. NSF CPS PI Meeting, Privacy-aware Medical Voice Applications, June 2021

T5. Baidu, Data-driven Security and Privacy analysis for emerging system, December 2020

T4. IMWUT/Ubicomp, VerHealth: Vetting Medical Voice Applications through Policy Enforcement, September 2020

T3. The Web Conference, Read Between the Lines: An Empirical Measurement of Sensitive Applications of Voice Personal Assistant System, April 2020

T2. NDSS, TKPERM: Cross-platform Permission Knowledge Transfer to Detect Overprivileged Third-party Applications, February 2020

T1. NSysS, Vulnerability Detection in Recent Android Apps: An Empirical Study, January 2017

Accepted CVEs

CVE-2021-41208 (Buffer Overflow, CWE-824, CVSS 9.3)
CVE-2022-21736 (Buffer Overflow, CWE-476, CVSS 7.6)
CVE-2022-21740 (Buffer Overflow, CWE-787, CVSS 7.6)
CVE-2022-21735 (Divide By Zero, CWE-369, CVSS 6.5)
CVE-2022-23567 (Integer Overflow, CWE-190, CVSS 6.5)
CVE-2022-21739 (Buffer Overflow, CWE-476, CVSS 6.5)
CVE-2022-23568 (Integer Overflow, CWE-190, CVSS 6.5)
CVE-2022-23569 (Buffer Overflow, CWE-617, CVSS 6.5)
CVE-2022-21734 (Buffer Overflow, CWE-843, CVSS 6.5)
CVE-2022-21737 (Buffer Overflow, CWE-754, CVSS 6.5)
CVE-2022-21738 (Integer Overflow, CWE-190, CVSS 6.5)
CVE-2021-29584 (Integer Overflow, CWE-190, CVSS 6.5)

Industry Experience

Research Intern in Baidu USA **June, 2021 - August, 2021**

- Developing GDPR compliant android mobile application → **Published one paper [IEEE CNS'22]**
- *DelChk*: Validating data deletion practice in websites

Worked with Dr. Ping Li and Prof. Yingjie Lao

Software Engineer in Kona SL **March, 2016 - July, 2017**

- *Kona Money*, a payment solution, developed both in Android and iOS platform

- *ArtMining*, a CMS based project, developed in iOS platform
- *Nexus Pay*, leading payment solution in Bangladesh, developed in iOS platform and ingenico POS terminal in C platform

Software Developer in Upwork [Profile Link] **January, 2013 - December, 2015**

- Successfully completed 12 jobs on developing Android apps